The Reaction to Separation in Infant Monkeys: Anaclitic Depression and Conservation-Withdrawal

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The reaction to removal of the mother was studied in 4 group-living pigtail monkey infants. All showed distress, with 3 progressing to a state of deep depression similar to the anaclitic depression of human infants following separation, as described by Spitz. The only infant not showing deep depression was the offspring of the dominant female. The stages of reaction are seen as successive efforts at adaptation based on available response systems, evolved for their selective advantage or developed ontogenetically, especially through dominance-hierarchical regulatory influences. In this regard the reactions have apparent survival value, in part through their communicative significance. Monkey infants have a greater chance of survival without a mother figure than humans because of their greater locomotor ability, which appeared to initiate recovery from the depressed state. The data support Engel's theory of two primitive biological response systems for handling distress, each with a mediating neural organization.

FOR HIGHER ORGANISMS who begin life with a long period in which independent functioning is not possible, normal development depends upon continuous parental care. Accordingly, considerable theoretical and practical interest has centered on the dramatic and deleterious effects often seen in young children who are separated for varying periods of time from the mother figure.[1,12,21,23,25,48,54,61,62]

The two observers who have most forcibly brought this problem to attention are Spitz and Bowlby. Spitz described two syndromes in institutionalized infants. In one,[62] which he called "anaclitic depression," the syndrome arose in the second half of the first year of life when the mother, who had raised the infant from birth, was removed for a time. After an initial period of apprehension and weepiness the infant showed

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Supported by Grants MH-01670, K3-MH-23,685, and 1-F3-MH-31,925 from the National Institute of Mental Health, U. S. Public Health Service.

A preliminary report of this study was published in *Science*, Feb. 24, 1967.

We wish to thank William Barnett for his very substantial contribution to this study, and also Drs. Glenorchy McBride, Walter Riss, and Albert Scheflen for their valuable and stimulating coments.

Received for publication Dec. 6, 1966.

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withdrawal, rejection of the environment, retardation of reaction to stimuli, slowness of movement, loss of appetite, increased finger sucking, insomnia, "an obvious distaste for assuming an erect posture or performing locomotion," and a "physiognomic expression . . . difficult to describe . . . [which] would, in an adult, be described as depression." Spitz said that their appearance prompted observers to ask if they were sick.* Over the course of this reaction the infants showed a retardation of development. If the mother returned within 3–5 months there was an immediate dramatic effect. "They suddenly were friendly, gay, approachable," and subsequently recovered fully. If she did not return, the other, more malignant syndrome developed. The latter,[61] which he called "hospitalism," was first seen in infants who lost their mothers permanently and were cared for by very busy nursing personnel. It was characterized by massive failure of development, both mental and physical, frequent illness, marasmus, cachexia, and often, early death. He commented on the importance of locomotion and motility after 6 months of age and felt that, among other things, the loss of the mother interfered with the opportunities for locomotion and its important role in development. In noting that not all the infants who lost their mothers after 6 months became depressed and that some became only mildly depressed, he attributed the variation to the nature of the mother-infant relationship, correlating the presence of depression with a good previous relationship—i.e., the good mother is missed. He did not define a good relationship, but it is reasonable to assume that at that age it was characterized by a high level of parent dependency in the infant. He strongly recommended that if the mother is lost a good substitute be provided, as well as ample opportunity for locomotion and motility.

Robertson and Bowlby[5, 48] described the reaction of children 2–3 years old who had to be hospitalized and thus separated from their mothers. The reaction was described as falling into three phases: (1) "protest"—in which the child cried and tried to regain the mother; (2) "despair"—in which the child became apathetic and withdrawn and didn't accept the ministrations of strangers; and (3) "detachment"—in which the child sought new relationships but became detached from and hostile to the mother, even after reunion.

These and all other studies of this problem in the human have necessarily been restricted to naturally occurring separations – e.g., hospitalization, and with the exception of Spitz's study,[62] in which only the mother was removed, all systematic studies have been confounded because the separation was not only from the mother figure but from the usual home site and family group as well.

Since experimental study of reactions to separation in the human is generally impossible, interest has developed in recent years in studying mother-infant separations in closely related forms, specifically, nonhuman primates. Even so, to our knowledge only 5 studies have been reported which specifically deal with reactions to separation.

Spence[60] separated infant chimpanzees from their mothers at various ages, but he studied the reactions of the mothers. They showed strong emotional reactions

*The similarities between depressive illness and physical illness were noted by Darwin,[11] who said, "The appearance of dejection in young orangs and chimpanzees, when out of health, is as plain and almost as pathetic as in the case of our children. This state of mind and body is shown by their listless movements, fallen countenances, dull eyes, and changed complexion." He also said, "After the mind has suffered from an acute paroxysm of grief, and the cause still continues, we fall into a state of low spirits; or we may be utterly cast down or dejected. Prolonged bodily pain, if not amounting to an agony, generally leads to the same state of mind." (See also Engel.[16])

and own-infant recognition even after 6 months.

Jensen and Tolman[51] studied 2 infant pigtails (*Macaca nemestrina*) aged 5 and 7 months, and their mothers, during separations of 5 min. in which babies were exchanged, and during the subsequent reunions. Each pair had lived in isolation before the study. The act of separation produced strong emotional responses, with agitation and continuous screaming by the infants. Both infants initially tried to approach and contact the strange mother, but these efforts waned quickly in the face of negative responses by the strange mothers. Upon reunion there were striking increases in positive behavior by both mothers and their infants (contact, clinging, cradling, grooming). It would seem that the mothers showed strong own-infant-specificity, whereas the infants were initially not own-mother specific but quickly became so after experiencing rejection and disregard by strangers. It should be noted that these infants had had no previous experience with animals other than their own mothers.

Two studies of separation have been done at the Wisconsin Primate Center. In the first,[58] 4 mother-infant pairs were denied physical contact for a 3-week period but were allowed auditory, visual, and olfactory communication. The infants were 6 and 7 lunar months old. Each pair had lived physically apart from but in visual and auditory contact with another dyad. Each infant had had some prior social experience with the other infant, who during the separation was also separated and available for social interaction at times. The initial reactions to separation were described as showing a "high degree of emotional disturbance," including disoriented scampering, screeching, and crying. During the separation period the infants showed increased crying and mother-viewing but decreased peer-directed social behavior and self-directed mouthing and sucking (in 3 out of 4). Immediately after reunion there were increases in mother-infant cradling, in infant-mother clinging, and in ventral-ventral contact, but the persistence of these effects varied among the pairs.

In their second study,[59] separation was total for 2 weeks. Again there were other separated infants, previously known, available for social interaction at various times. The infants were 7 lunar months old. Eight pairs were studied. The initial reactions to separation were much as in the earlier study. During separation crying increased significantly, and self-mouthing decreased in 6 out of 8. There was a significant decrease in complex peer-directed social behavior and infant-infant oral behavior, but a significant increase in infant-infant nonspecific contact. There was some aggressive behavior during and after separation, and after reunion a significant increase in several play behaviors which seem to be part also of aggressive encounters. On the first day following reunion there was a significant increase in clinging by infants and cradling by mothers, but this increase disappeared when the entire 2-week reunion period was compared with the 2-week pre-separation period, which prompted Seay and Harlow to state, "The result of mother-infant separation on the mother-infant relationship of these animals was transient and apparently unimportant. This does not indicate that the effects of repeated or prolonged separations would not seriously disrupt the infant-mother relationship." However, concerning the reactions during separation they cautiously concluded that the severe "depression of play and other complex social behavior is indicative of severe emotional *distress*." (our italics). They also noted two stages in the reaction. After describing the initial stage as "violent protest," they said, "They then passed into a stage characterized by low activity, little or no play, and occasional crying. The writers

feel this second stage is behaviorally similar to that described as despair of children separated from their mothers." However, they gave no further description of this stage aside from the statistical data already noted.

Finally, Hinde *et al.*[28] reported on the effects of separation in infants who had been living in groups consisting of a male, 2 or 3 females, and their offspring. There were 4 rhesus infants separated between 30 and 32 weeks of age for a period of 6 days. Upon removal of the mother there was an immediate increase in distress calls, which then decreased during the course of the separation, but was still at a higher than preseparation level 4 weeks after the mother's return. During the separation the infants directed behavior toward other adults which they had previously directed toward their mothers, but they received much less care from the other adults than they had from their mothers. Whereas they had been on their mothers in 41–53% of observation periods, they were on other adults during separation only 0–24% (mean, 17%). During separation there was a fall of general activity (locomotor behavior), as well as of social and manipulative play. All three behaviors showed some increase during the course of separation, and in three of the four returned to preseparation levels within a week of the mother's return.

On the first day of return all 4 infants spent a great deal of time on the mother, but there was individual variation which was directly related to preseparation levels—i.e., the more they had been off the mother before separation, the less they clung when she returned. For 2 infants preseparation levels were reached by a week after return, but in the other 2 the effect was longer-lasting, in 1 being manifest even 20 weeks later. In view of the over-all reduction in activity, the authors state that during the separation "the behavior of the infants can be described as *depressed.* (our italics). They usually sat in a characteristic hunched posture. . . ." The authors concluded that these rhesus infants were adversely affected by the separation experience, and that the severity of the effect varied with the nature of the previous mother-infant relationship. They also concluded that the rhesus infants showed symptoms which in many ways parallel those of human infants undergoing separation.

Both Seay *et al.* and Hinde *et al.* compared the reaction of rhesus infants to the human reaction described by Bowlby, and concluded that there is similarity with respect to the stages of "protest" and "despair" but that "detachment" with hostility to the mother did not occur in the monkeys. The question of what stages of reaction appear in the infant separated from the mother, however, is by no means completely consistent in reports on humans. It should be recalled that Spitz found only two stages—i.e., apprehensive crying followed by depression; he did not find hostility to the mother and "detachment" from her.

Likewise Schaffer and Callender[53, 54] in a study of hospitalized 7- to 12-month-old infants found evidence of anxiety and depression but not detachment from the mother. On the other hand, Heinicke and Westheimer[25] in a detailed study of 2- to 3-year-old children in a residential nursery delineated five phases of psychic reaction, which they saw as "successive efforts at adaptation." It seems likely that the differences reported among human children are to be explained at least in part by the difference in age: 7–12 months in the one study, and 2–3 years in the other. Another age difference was described by Schaffer and Callender concerning hospitalized infants under 7 months of age. In the face of good care by hospital personnel—i.e., adequate substitute mothering, these younger infants did not show anxiety or depression, although upon return home they showed an enormous preoccupation

with the environment. In the first 6 months it seems that the infant requires mothering but that "mother" as such has not yet been sharply delimited, whereas afterwards "mother" per se has specific meaning. This idea is supported also by Rheingold's observation[44] that 6-month-olds in an institution became more socially responsive when cared for primarily by one person, rather than by a number of nurses.

It is obvious that in making comparisons, even within a species, consideration of levels of development is crucial. The same caution applies to interspecies comparisons. For this reason we shall not deal in this report with elaborations of psychic function, or address ourselves to previous theoretical disagreements.[4, 20, 57, 63] We are concerned here with the more primitive aspects of the reaction, with our evolutionary heritage, so to speak.

In a theoretical paper, Frank[19] offered "the thesis that depression-elation responses constitute part of the inherent adaptive machinery available to the individual. They are employed automatically, unconsciously, and directly as adaptive measures under conditions when either in actuality or in fantasy a relatively helpless individual is threatened with the loss of suitable care, protection and sustenance."

Engel and Reichsman,[14] in their classical study of the child Monica, described how she would withdraw in the presence of a stranger and even go to sleep, while showing profound reduction in the secretion of hydrochloric acid by the stomach. Drawing upon the thesis offered by Frank, they labeled Monica's response as "depression-withdrawal"; "depression" because of "the impact on the observer of the facial expression, posture, and inactivity, all of which call to mind a mood of dejection, sadness, or depression"; and "withdrawal" because of "the turning away, the closing of the eyes, and the eventual sleep. The immobility and hypotonia may be seen as part of the withdrawal pattern as well as part of the affect disturbance." It seemed to them that Monica showed elements of both anaclitic depression and hospitalism as described by Spitz.

In subsequent publications Engel[16, 17] has elaborated this thesis further. Incorporating observations on infants and young children[7, 24, 45] and the theorizing of Bibring,[3] he wrote, "When we couple such observations with the recognition of the fact that even the most intense crying fit eventually terminates and the infant becomes quiescent, falls asleep, or even becomes comatose, even though the underlying needs have not been fulfilled, we are drawn to the conclusion that the central nervous system is organized to mediate two opposite patterns of response to a mounting need. . . . The first, broadly subsumed under the heading of flight-fight, involves activity, energy expenditure and engagement with the environment to control sources of supply and avert danger. The second, conservation-withdrawal, involves inactivity, energy conservation, raising of the stimulus barrier, and withdrawal from the environment. Each of these is seen as an inborn system, each with its own underlying mediating neural organization. The conservation-withdrawal system is a second biological defense organization which comes into play if and when the energy expenditures of the first (flight-fight) reaction threaten the organism with exhaustion before supplies are secured. . . ." He considers the first system to be the biological anlage of anxiety and the second of depression-withdrawal, these being in his view the two primary affects of unpleasure. He suggested that neurophysiological data on the diencephalon and limbic system supported the view that these two systems have inborn mediating neural organizations. Together with Schmale,[18] he has also proposed that the derivative affects of depression-withdrawal are importantly

involved in the development of somatic illness, through their frequent activation by object loss. Schur,[57] in commenting on the biological precursors of the ego reactions of anxiety and depression, also asserted their importance in psychosomatic reactions, as has Reiser[43] in a recent proposal.

The present report will offer support for the thesis that the stages of reaction to separation represent successive efforts at adaptation based on available response systems evolved for their selective advantage. The communicative value, primitive emotional aspects, and underlying neural mechanisms of these responses will also be discussed. In addition, the ontogenetic influence of the regulation of monkey behavior by the dominance hierarchy will be considered, especially as it affects the reaction to separation. This report is based on a study of separated pigtail monkey infants which is part of a broad ongoing comparative program of research on infant development and mother-infant relations in group-living monkeys of several species. This research followed 4 years of normative studies.[35, 51]

Subjects and Methods

The subjects were 4 pigtail (*Macaca nemestrina*) infants, 4.8–6.1 lunar months of age (mean, 5.6) when separated, born in the laboratory and raised from birth by their feral mothers in a group with their sire and an infantless adult female (Table 1). The group, which had lived together for over 2 years at the start of this study and had stable hierarchical relations, was housed in a pen, schematized in Fig. 1, which was 8 ft. wide, 13 ft. deep, and 7 ft. high. All observations were made by one of two experienced observers, through two one-way vision windows in the front of the pen. The monkeys could not see outside their pen.

The physical separation of mother and infant was done outside the pen with a minimum of handling and trauma. The mother was placed in a separate pen, and the infant was returned to the home pen with the group in which it was born and reared. After 4 weeks the mother was brought back and released into the pen containing her infant and the rest of her group. There was a 3-week overlap in the separations of the first 2 infants studied, and again in the separations of the second 2 infants studied approximately 2½ months later (see Table 1).

Observations were started 1–2 months before separation and continued for 3 months after reunion. They were spread over 3–5 days per week, totaling approximately 2 hr. of observation per week, for each dyad in the month before and after separation, and for each infant alone during the separation, and for about half that time in the second and third months after reunion.

Observations were recorded with a keyboard device connected to a series of clocks and counters. This enabled the observer of each session to record the total duration and frequency of occurrence of a wide variety of behaviors by the mother and/or the infant. The behaviors scored were those, selected from the total repertoire of these animals as listed and defined in our taxonomy,[34] which seemed relevant to this study. They included for the infant, both filial and non-

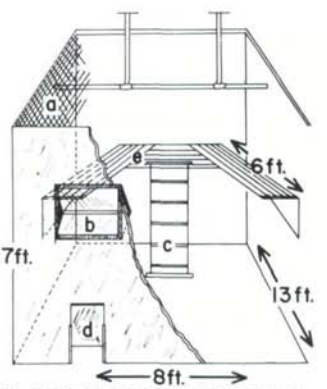


FIG. 1. Scale diagram of group pens: *a*, overhead mesh; *b*, one-way vision screen; *c*, mesh-covered ramp; *d*, guillotine animal door; *e*, aluminum shelves.

TABLE 1. IDENTIFYING DATA ON SEPARATED INFANTS AND THEIR MOTHERS

Infant	Sex	Birth date	Age at separation		Age of other infants at time of separation								Mother's rank in female hierarchy
					A		B		C		D		
			Days	Mo.*	Days	Mo.*	Days	Mo.*	Days	Mo.*	Days	Mo.*	
A	Female	1/10/65	171	6.1	—	—	145	5.2	53	1.9	15	0.5	1
B	Male	2/ 5/65	152	5.4	178	6.4	—	—	60	2.1	22	0.8	2
C	Male	5/ 8/65	164	5.9	282	10.1	256	9.1	—	—	126	4.5	4
D	Male	6/15/65	133	4.8	289	10.3	263	9.4	171	6.1	—	—	3

At the time of separation for B, A had already been separated for 7 days; similarly, at the time of separation for D, C had been separated for 7 days.

* Lunar.

filial behaviors, and for the mother both maternal and nonmaternal behaviors. In addition, at the end of each session the observer's subjective impressions were recorded. Prior to separation and following reunion, each observational session was divided into three sections which focused observations as follows: (1) 500 sec. (8.3 min.) on dyadic interactions—i.e., maternal and filial behaviors; (2) 250 sec. (4.2 min.) on nonfilial infant behaviors, and (3) 125 sec. (2.1 min.) on nonmaternal behaviors by the mother. During the separation itself, the infant was observed for 500 sec. (8.3 min.) during each session. The observational sessions were replicated several times each observations as follows: (1) 500 sec. (8.3 min.) into percentages of total observation time, to equate for variations in duration of testing. Repeated measures analyses of variance were utilized throughout to assess the statistical significance of the data obtained.

Results and Discussion

Course of the Reaction

The immediate reaction to physical separation of mother and infant, as noted by all previous observers, consisted of loud screams by both, and massive struggling to regain each other. When the infant was replaced in the pen he was highly agitated (Fig. 2). Pacing, searching head movements, frequent trips to the front door and windows, sporadic and short-lived bursts of erratic play, and brief movements towards other members of the group seemed constant. Cooing, the rather plaintive distress call of the young macaque, and intermittent screeching were frequent. This reaction persisted throughout the first day, during which time the infant did not sleep. All observers were struck by a sense of intense acute distress.

After 24–36 hr. the pattern changed strikingly in 3 of the 4 infants, B, C, and D. The infant sat hunched over, almost rolled into a ball (Fig. 3) with his head often down between his legs (Fig. 4).

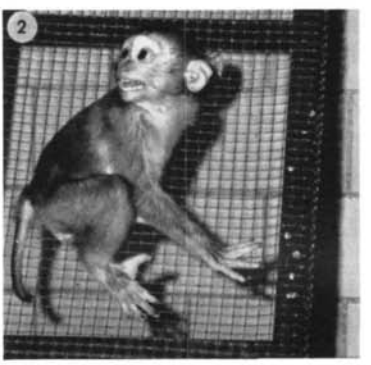


FIG. 2. Agitated pigtail infant fleeing from rebuff by adult female he had approached. Note fear grimace while screaming.

This position for a time concealed from us the fact that 2 of the infants, C and D, were sucking their own penis.* When the face could be seen it seemed that the facial muscles had sagged, which, together with the configuration of forehead and mouth, created the same appearance of dejection and sadness (Fig. 5) which Darwin[11] described and believed "to be universally and instantly recognized as that of grief." Movement virtually ceased except when the infant was actively dis-

*This is an unusual behavior in our laboratory, having been seen only once before, in an 18-month-old pigtail at the time a sibling was born and he was temporarily displaced from his mother. This infant had been separated from his mother for 8 days at 11 months of age, during which time he too was severely depressed.

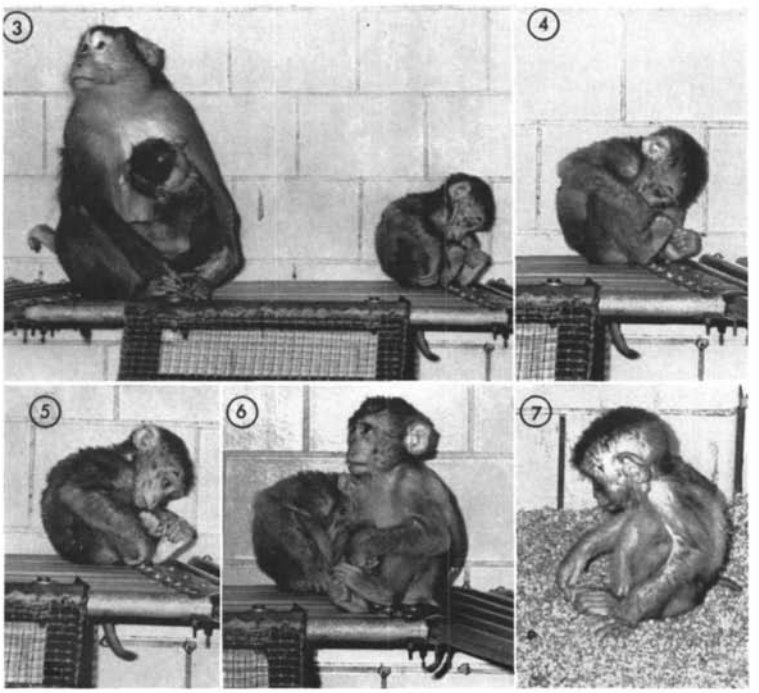


FIG. 3. Depressed pigtail infant showing characteristic hunched-over posture with flexion throughout. He is completely disengaged from mother and infant nearby in ventral-ventral contact. FIG. 4. Depressed pigtail infant showing characteristic posture including head between legs. Note slightly opened eyes as he sucks his penis. FIG. 5. Depressed pigtail infant showing characteristic posture and dejected facies. FIG. 6. Two depressed pigtail infants. The one in rear shows characteristic hunched-over posture. The one in front has lifted his head to look across pen. Despite their passive contact they are quite disengaged from each other. FIG. 7. Depressed infant showing aimless, tentative exploration of bedding during early stages of recovery.

placed. The movement that did occur appeared to be in slow motion, except that movement could be quick at feeding time or in response to aggressive behavior. The infant appeared to be moping much of the time and rarely responded to social invitation. Also, he rarely made a social gesture, and play behavior virtually ceased. He seemed largely disengaged from the environment and appeared to derive little comfort from the presence of others (Fig. 3 and 6). (There was a significant rise in contact with other infants during the separation month ($p < 0.05$), but the rise was especially marked in the last half of the month when recovery was proceeding. It was rarely ventral-ventral contact.) Occasionally the infant would look up and plaintively "coo." To all observers, including many visitors, there was communicated a strong feeling of dejection, withdrawal, and social isolation. We can only characterize the infant's reaction as the expression of a state of severe depression.[11] The coat often looked shaggy but no obvious physical illness occurred, although observers often commented that the infant looked sick. Eating was generally adequate and at times voracious.

After persisting unchanged for 5–6 days the depression gradually began to lift. The recovery started with the resumption of a more upright posture and a resurgence of interest in the inanimate environment. Slow tentative exploration appeared with increasing frequency (Fig. 7). Gradually, the motherless infant also began to interact with his social environment, primarily with his peers (Fig. 8), and there was then a reemergence of play activity. The depression continued, but in an abated form. Periods of depression alternated with periods of inanimate object exploration and play. By the third and fourth weeks of separation, play levels were approaching those of preseparation. Movement increased in amount and tempo. Toward the end of the month the infant appeared alert and active a great deal of the time; yet he was still not a typical infant of that age, as indicated both by the still-present periods of depression and the various behavioral indices.

Another dramatic change appeared when the mother was reintroduced to the group. Physical reunion via ventral-ventral contact was intense and lasting (Fig. 9), to the extent that inanimate object exploration and play, both of which had risen to near-preseparation levels by the end of the separation month, fell during the reunion month to even lower than preseparation levels (in all 4 subjects).

Major Behavioral Components in Common

Despite individual variation there were certain significant changes which characterized the entire group. These are best seen by a comparison of the month before separation with the separation month and the 2 months subsequent to reunion with the mother. Digit suck, self-orality (mouthing or licking other parts of the body) and genital self-stimulation (Fig. 10), all of which reflect stimulation of the self, when combined, showed a significant increase (from 0.9 to 1.8%, $p < 0.05$) during the separation month. Following reunion, there was a marked decrease to a level (0.35%) which was still below preseparation in the second month following reunion (0.6%). Each infant showed a rise during separation; it is interesting to note that in 2 the major part of this increase occurred on the first day of separation, whereas in the other 2 infants the rise occurred later in the separation month (Fig. 11). Our results differ from the Wisconsin studies on rhesus infants in this regard and are more like what Spitz reported for separated human infants. It is unclear whether species differences alone account for the differential findings in this regard.

The separated infants also showed a

significant increase ($p < 0.01$) in the amount of inanimate object exploration. This behavior too showed a marked decrease following reunion to levels below that of the preseparation month (Fig. 12). The greatest rise in inanimate object exploration was seen in infant A, the only one who did *not* get depressed.

Further emphasizing the isolated quality of the separated infants' behavior was the drastic fall in play, both social—e.g., rough-and-tumble or chasing (Fig.

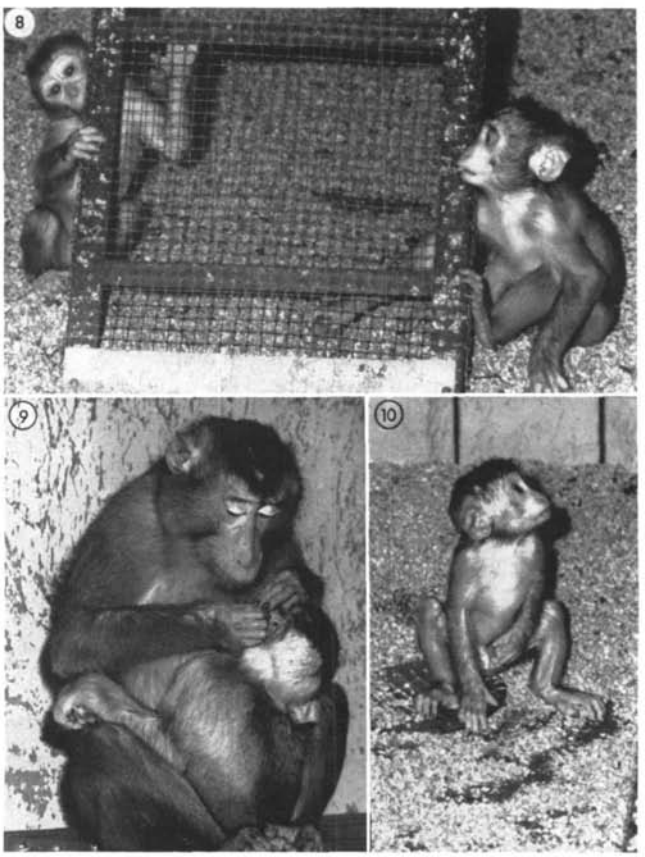


FIG. 8. Two depressed infants separately exploring inanimate environment. One is sucking edge of ladder; other leans on ladder while handling bedding. FIG. 9. Reunited mother and infant in ventral-ventral contact. Mother is grooming infant's ear. FIG. 10. Separated pigtail infant showing genital self-stimulation.

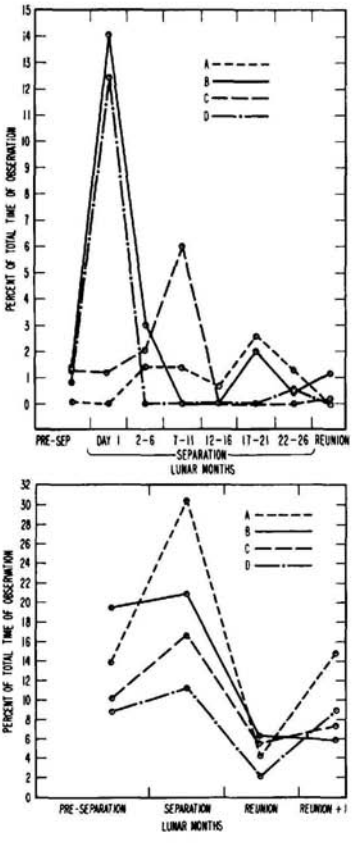


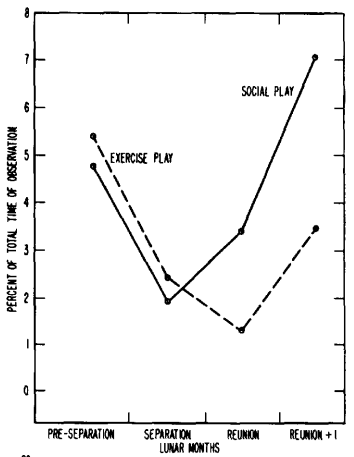
FIG. 11 *(top)*. Mean duration for each infant of digit suck, self-orality, and genital self-stimulation combined, during preseparation month, successive parts of separation month, and reunion month. All months were calculated on lunar basis (28 days). With exception of Day 1, separation month is divided into 5-day segments; Days 27 and 28 are omitted. FIG. 12 *(bottom)*. Mean duration for each infant of inanimate object exploration, during preseparation month, separation month, and 2 months following reunion.



FIG. 13. Two nonseparated pigtail infants clinging to wire mesh ceiling while engaged in social play (wrestling). Third infant nearby is exploring bar attachment by sucking, looking, and handling.

13) and solitary (exercise) during the separation month ($p < 0.01$) (Fig. 14). This decrease is particularly striking since normally during this age period play behaviors are expanding quite rapidly.[35] For both kinds of play, each infant showed a marked decrease for the separation month, although the levels of these behaviors began to rise later in the month (Fig. 15 and 16).

There were some striking effects noted following reunion, in the form of an enormous reassertion of the dyadic relationship, with marked increases in various measures of closeness in all 4 reunited pairs, as compared to preseparation levels. Clinging by the infant, protective enclosure by the mother, and nipple contact all rose significantly ($p < 0.01$). Even in the second month following reunion this trend was still evident. This significant rise in measures of dyadic closeness is quite striking when we realize that ordinarily for the age periods involved these particular behaviors show a fall of one third to one half.[35] The increased closeness was manifested in other



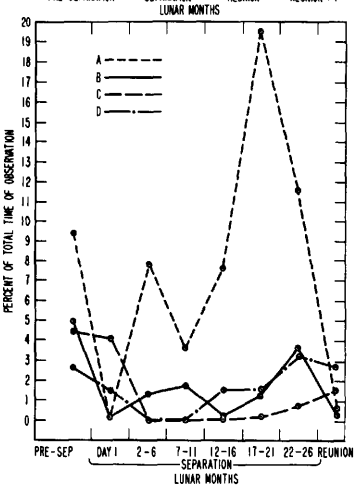


FIG. 14 *(top)*. Mean duration for all 4 infants of social play and exercise play, during preseparation month, separation month, and 2 months following reunion. FIG. 15 *(bottom)*. Mean duration for each infant of exercise play, during preseparation month, successive periods of separation month, and reunion month.

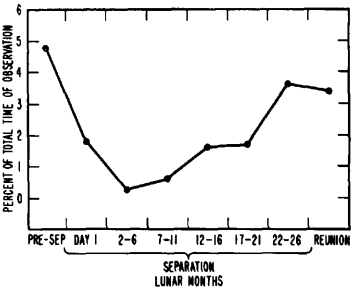


FIG. 16. Mean duration for all 4 infants of social play, during preseparation month, successive periods of separation month, and reunion month.

ways as well. A measure of mother-infant physical separation that we have found valuable in our normative studies concerns departures by the infant or mother from the other (ordinarily by the infant) to another level of the pen. The frequency of such other-level departures fell by 80%. Furthermore, the mean bout length of these other-level departures fell from 60.5 sec. in the month before separation to 34.4 sec. in the month following reunion.

Finally, in this context, maternal behaviors that normally discourage dyadic cohesiveness at this age, such as punitive deterrence (e.g., gentle biting of the infant's body) and nipple withdrawal, appeared very rarely. Thus our results concerning the effects of the separation on the mother-infant relationship differ dramatically from those of Seay and Harlow[59] on rhesus monkeys. The separation for 4 weeks certainly had a lasting influence on the mother-infant relationship. Whether this difference is due to the species difference or to the longer period of separation we used in our study we cannot say. However, since their dyads each lived alone in adjacent but separate cages it is likely that dominance-hierarchical pressures were minimal, and this would account for a gen-

eral reduction in maternal protectiveness and infant dependency, as compared to a group-living situation.

Adaptation and the Problem of Survival

Our observation of major behavioral changes in the infant in response to the loss of the mother is concordant with the other evidence. Various observations and reviews[6, 9, 11, 40, 65] have documented the occurrence of severe reactions in animals to the loss of close objects, especially in infants to the loss of mother. Obviously the loss of mother poses a serious problem for the survival of the infant. However, for nonhuman primates there are only meager data on what ultimately happens to these separated infants under natural conditions. For the purpose of this discussion we exclude from consideration the important studies, especially those by Harlow, of infants reared in almost total isolation and stimulus deprivation in the laboratory.

First of all, there would appear to be great survival value in a mechanism which provided a substitute caretaker, as by adoption. Zuckerman[66] reported that a young female baboon became very close to a "bachelor" after the death of her mother. He later became her only companion. DeVore[13] reported an apparent adoption by a beta male baboon of a sick female infant. When observations began he noted 2 very emaciated adult females, who then disappeared within a few days. "One of these females was *apparently* (our italics) the sick infant's mother." The male protected the infant who became his constant companion. Rowell *et al.*[52] reported that an 8-month-old rhesus infant whose mother died was adopted by an infantless "aunt" who had been a close companion of the mother. They pointed out the obvious survival value of such an adoption, and stressed the previous close relationship between mother and "aunt." Jay[30] reported that she had observed a bonnet female carrying 2 young infants, and speculated that this was an adoption since another bonnet female had been killed several days before.

In none of these reports were any details given concerning the behavior by the infant which might have led to the adoption; nor was the behavior described of the other animals in the setting just prior to the adoption. Yet both seem important as determinants of whether an adoption will take place.

In our own experimental studies with bonnets we have seen numerous adoptions, but this will form the basis for a separate report.

Individual Variation and the Role of Hierarchical Factors

There is conflicting evidence concerning a youngster's behavior in the matter of finding a substitute for a lost mother. Freud and Dann,[22] in their study of concentration camp children, found that the children used whomsoever they could to relate to, in that case to each other. The study by Jensen and Tolman[31] indicated that a pigtail infant separated from its mother will attempt to relate to a strange mother, and stops only when rebuked. On the other hand, it is well known that children separated from their mothers often refuse the ministrations of strangers. It seems likely that these differences are due to a number of factors, including: (1) differences in developmental levels; (2) their being manifestations of different stages of the reaction to separation; (3) differences in the contextual situation; and (4) individual variation based on genetic and experiential, including sociocultural, determinants.

There are at least two components involved in successful turning to a substitute. The first is the effort by the infant to do so, and the second is the nature of the adult responses. Both of these are likely to be modified by previous experience. The adoption cited by Rowell *et*

al.[52] was made by an adult who had had a previous positive relationship to the infant and the infant's mother. (The other reports offered no data on this matter.) This would suggest that an adult who had a close relationship to an infant and his mother would be more likely to respond positively to an infant's appeal for caretaking.

In monkeys the likelihood of a previous positive relationship between a young infant and an adult other than the mother depends in large part on the hierarchical status relationships between the infant's mother and the other animal. In general the evidence seems to be that an infant is more likely to have had a positive relationship with a female of the same or lower status than his mother. Rowell and Hinde[27, 52] reported that in the rhesus, mothers allowed subordinate females to carry and cuddle their infants, and this was especially true if the other female had been their usual sitting and grooming companion. On the other hand, advances to an infant, whether aggressive or otherwise, by a more dominant female were likely to produce protective behaviors by the mother, including the termination of the infant's interaction with the "aunt." They also reported that adult females would at times show aggressive behavior to infants not their own, and this was greater if they were dominant animals, and/or if they had an infant of their own.

DeVore[13] has pointed out that among baboons the offspring of low-ranking females tended to show much more insecurity, in the form of greater frequency of alarm cries and more demands on the mother, leading to an intensification of the mother-infant relationship (high parent dependency). He also commented that infants were responsive to the emotional state of the mother, so that when the mother was disturbed the infant ran to her. A low-ranking female continually subjected to dominance by others suffers more emotional upsets, bringing her infant to her more often; DeVore postulated that it is very upsetting to the infant to have his mother repeatedly distressed and demeaned. The effect here too is an intensification of the mother-infant relationship in the context of stress and insecurity. On the other hand, the offspring of the more dominant females were reported by DeVore to be much more secure and to have a greater freedom of movement. He cited a similar finding in the Japanese macaque: that the offspring of dominant animals themselves tended to be more domineering.

From this evidence we would surmise that the infants of more dominant females tend to be more secure, to have greater freedom of movement, to have less dependence on their mothers, to have had more varied and greater experience, and to have built up a greater repertoire of behavior than the infants of low-ranking females. We would thus expect that if the mother were lost the offspring of the dominant females would be freer in approach to other females, and would have a greater variety of behavior for coping with the situation, so that the experience would be less shattering for them. On the other hand, we would expect that the offspring of submissive females would not so freely approach other females, and by comparison would have an impoverished repertoire of coping behaviors, so that the experience would be overwhelming.

With regard to the adult females, we would expect from the evidence cited that the dominant ones, especially if they had infants of their own, would be more aggressive and less caretaking toward motherless infants, whereas submissive females, who are likely to have had friendly relations with other infants, would be expected to be less aggressive and more accepting. The sum total of these expectancies is that the offspring of submissive females have a greater risk of failure to survive. In terms of

TABLE 2. CHANGES IN BEHAVIOR BY INFANTS DURING FIRST 6 DAYS OF SEPARATION
(% OF TOTAL TIME OF OBSERVATION)

	Infant and time period											
	A			*B*			*C*			*D*		
Behavior	*Pre-sep. month*	*Day 1*	*Days 2–6*	*Pre-sep. month*	*Day 1*	*Days 2–6*	*Pre-sep. month*	*Day 1*	*Days 2–6*	*Pre-sep. month*	*Day 1*	*Days 2–6*
Digit suck, self-orality	0.1	0	1.4	0.6	14.1	3.0	1.1	1.2	1.1	0.2	7.2	0
Autogroom	0.8	0.4	2.0	0.6	0	3.6	1.1	0.1	0.7	2.1	1.2	1.3
Genital self-stimulation	0	0	0	0.8	0	0	0.2	0	1.0	0.6	5.4	0
TOTAL ACTIVITY TO SELF	0.9	0.4	3.4	2.0	14.1	6.6	2.4	1.3	2.8	2.9	13.8	1.3
Exercise play	9.3	0.1	7.8	5.0	0.2	1.2	4.4	4.1	0	2.7	1.4	0
Inanimate obj. exploration	13.8	8.4	25.2	19.5	4.0	43.5	10.0	17.9	4.0	8.8	20.9	1.9
TOTAL ACTIVITY BY SELF	23.1	8.5	33.0	24.5	4.2	44.7	14.4	22.0	4.0	11.5	22.3	1.9
Contact with adults	0.3	24.5	1.6	0.8	2.6	4.4	0.8	0.5	0.4	0	1.2	0.1
Pos. behavior to adults*	5.6	0.4	1.7	0.8	0	0	0.3	0.5	0.7	0.8	0.1	0
Pos. behavior from adults†	0	10.0	13.0	0	0.1	0.2	0.3	0.2	0.6	0	0.2	0
Neg. behavior from adults‡	0.3	0.7	0.8	0.1	0.1	0.5	0.1	0.1	0.1	0	0	0
TOTAL INTERACTION WITH ADULTS	6.2	35.6	17.1	1.7	2.8	5.1	1.5	1.3	1.8	0.8	1.5	.1

* Combined score for *social groom of*, *climb on*, and *return to* adults.
† Combined score for *social groom* by, *social exploration* by, and *retrieval* by adults.
‡ Combined score for *threat* from, *punitive deterrence* by, and *lipsmacking* and *present* to adults.

selective advantage of dominance, this is consistent.

Our study was carried out in a group which had lived together for a long time and had stable and known hierarchical relationships. Every infant had lived in the pen since birth and presumably knew every other animal present and available during the mother's absence. This allowed us to examine the data in ways not possible in previously reported studies.

In examining our data, it would appear that many of these expectations were confirmed. Although all the infants showed evidence of disturbance in response to the separation, as measured by the various behavioral indices, there were differences in coping and emotional reactions. Infant A, whose mother was the dominant female, showed the least disturbance, whereas infants C and D, whose mothers were the most submissive females, showed the greatest disturbance, and infant B, whose mother was the second most dominant female, showed an intermediate reaction.

As noted earlier, all the infants showed distress upon being separated, in the form of crying and screaming, pacing, and searching, and all showed a marked reduction of both social and exercise play throughout the first day of separation. However, whereas 3 became depressed the next day, 1 did not: A, the offspring of the dominant female. Since the period of maximum depression was from the second through the sixth days of separation, it is worth comparing the behavior of A, the nondepressed infant, with C and D, the most depressed infants, and finally to consider B, the least depressed infant, both on Day 1 and Days 2 through 6 of separation (Table 2).

Certain of their activities have been grouped into 3 categories: activity to self, activity by self, and interaction with adults. Whereas before the separation, interaction with adults did not occupy a great deal of the time of any infant, it was greatest by far in A. On the first day of separation there was a trend to increased interaction with adults, but it was most marked and very striking in A, who spent more than a third of her time in interaction with adults. The major part of this interaction was contact, and it should be noted that physical contact between pigtails in our laboratory has been an uncommon behavior[50] except between a mother and her infant. There was a rise in negative behavior by adults to A, but this was slight compared with the massive rise in positive behavior.

Concerning the activities directed to the self, 2 of the 3 later depressed infants showed a marked rise on the first day. With respect to activities by self, we see that exercise play fell in all 4 infants on the first day, although least in the 2 infants who later became most depressed. There was also a marked fall in inanimate object exploration in A (the nondepressed) and B (the least depressed) on the first day, whereas C and D showed a marked rise. (It is necessary to add a qualitative comment. The inanimate object exploration during this first day of separation was primarily an aimless poking in the bedding of the pen.) Thus, it appears that the coping by C and D on the day of separation was primarily through self-activity, whereas A turned to other adults.

During the next 5 days, however, C and D, who were then deep in their depressive reaction, showed a marked fall in inanimate object exploration and no exercise play, whereas A, following the initial depression in activity by self, during the time that she was busy with adults showed a marked rise of both inanimate object exploration and exercise play. B showed a trend similar to A's, with a rise in both these activities, but this was primarily in inanimate object exploration, which occupied him during this period over 40% of the time observed.

The data thus appear to be in line with our expectations. The offspring of the dominant female showed a freedom to interact with adults in the absence of her mother, which undoubtedly was comfort-producing and ameliorative of the effects of the mother's absence. For their part, the other females were more accepting of the alpha female's offspring and directed considerable positive behavior to that infant.* With this degree of comfort and security, the infant was able to restore a high level of motility in exercise play and a considerable involvement with the environment in inanimate object exploration. She remained engaged with her environment. She did not become depressed.

The infants of the low-ranking females apparently did not feel secure enough to approach other adults very much, although they tended to try. However, as might be expected, they did not succeed in eliciting positive behavior from the more dominant females. In fact they experienced more intense negative behavior, which is not reflected in Table 2. For example it will be noted in several of the illustrations that the depressed infants have minor flesh wounds on forehead and face. These injuries were done one night during the time that C and D were both depressed while their mothers were out of the pen. It was the fourth night of separation for D, the eleventh for C. When we arrived in the morning we found B, C, and D all cut about the face. We assume that the damage was done by the mother of A. With failure to achieve a comfort-producing interaction with adults, C and D became depressed.

Infant B, who fell in between, did show the depressive reaction with a temporary disengagement from the social environment but not from the inanimate environment. The modest rise in interaction with adults he showed in the first 6 days reached major proportions in the second week of his separation, during which time he showed a major lifting of the depressive reaction. He was seen to sleep alongside of and in contact with the delta female.

On reunion, there was also individual variation in behavior. The dominant female and her infant approached each other immediately and effected close ventral-ventral contact. The other 3 mothers did not immediately go to their infants but instead approached the male and the other females, giving a variety of submissive or dominant signals—i.e., reestablishing their hierarchical status. This was especially true of the delta female, and when she finally took her infant, he stayed briefly and then departed from her for a time before they reunited more permanently. During the period that the mothers were busy reestablishing their hierarchical relationships, Infants C and D were continuously making approaches to the mother, which were unsuccessful until she finished with the other adults. Only Infant B did not immediately approach his mother. In this reunion behavior we see additional evidence of the hierarchical regulation of monkey behavior.

Adaptive Significance of the Stages of Reaction

To consider further the problem of survival each stage of the reaction to separation may be examined. The young of every species which is not independent at the beginning of its existence has been born or hatched preadapted to early life in many ways, including having behaviors which are calculated (by evo-

*We realize that the validity of this difference is confounded by the fact that A was the only female infant. However, it should be noted that in the Japanese macaque, caretaking of temporarily motherless infants does not discriminate sex at 1 year of age. (See the footnote on page 666.) Also, all available evidence from field and laboratory indicates that male infants are more active and independent than females, which suggests that from sex differences alone we would expect male infants to have greater coping ability than females.

lutionary adaptation) to maintain closeness to the mothering figure. These are frequently vocal. The eliciting causes of these behaviors in the beginning are varied and relate to the infant's need for dry warmth, food, etc.* The monkey infant has at least two calls which are distress signals, the scream and the "coo." These location calls match reciprocal mechanisms in the mother to ensure her closeness and to increase the likelihood of the infant's survival.

In this study, distress calls were quite frequent in the first stage following separation. In the wild, had the infant become separated from the mother, the calls would surely have had the effect of bringing her back, were she able. The monkey infants also showed restless pacing and searching which, in the wild, would have been over a wider area and most likely would have assisted in reuniting the infant with his mother. However, this calling, searching reaction is not the kind of mechanism that could profitably go on indefinitely, for at least two reasons. First of all, the increased movement and vocalization increase the risk of provoking aggression, either from conspecifics or from predators. Secondly, the markedly heightened activity is exhausting. Thus, we find that the agitated distress reaction comes to an end after a relatively short period of time. (Apparently it may also become extinguished or in part stereotyped if not reinforced, since it is not seen in the same form in chronically distressed animals such as long-term isolates.)

The second stage of the reaction to separation in 3 of our 4 monkey infants we have described as a severe depression. There appears to be survival value in this also. First of all, there is an obvious conservation of energy and resources. This is not depletion of energy or massive fatigue, because under certain circumstances the infant moved quite rapidly and appropriately—e.g., when food was put into the pen, or when an act of aggression was directed at him. Conservation is achieved through the marked inactivity and the posture (body rolled almost into a ball with flexion throughout, including the head between the legs) which likely cuts down heat loss by reducing the exposed body surface. The posture also makes the infant literally smaller, so that he presents less of a visual stimulus to other animals. The paucity and extreme slowness of movement also minimize notice of him by others. He is thus less likely to provoke aggression. Furthermore, the posture would reduce exposure to adversive elements—e.g., wind and rain. Finally, the reduction in exposure of body surface, and the hiding of the face, markedly reduce sensory input to the infant, the significance of which will be discussed below. Let it here be noted only that ordinarily in a social situation, a monkey is apparently able to avoid responding to a communication if he avoids receiving it.

From the communicative aspect, the depressed infant receives few signals and puts out few likely to provoke aggression. It also seems possible that the configuration, "dejected baby," or "sick baby"—the collapsed posture, the paucity and slowness of movement, the "coos" and the facies—might have specific signal value. Ordinarily, infants signal their distress with first-stage behaviors such as screaming, crying, and searching, which invariably bring about responses

*A chick has a distress call, for example, which is elicited whenever it is cold or wet or without food. It is also a location call which has the effect of bringing the hen to the chick since the hen has a fitting reciprocal mechanism. Schleidt[56] has shown that turkey hens who have been deafened as chicks will breed normally but will not only not respond positively to the distressed and calling chick, but will peck it as they might a strange object, even to death. Ordinarily the reciprocal mechanisms work quite satisfactorily to ensure the chick's survival. Even in the domestic chick, however, social bonding occurs early in life, and the disruption of the social bond then becomes in itself a sufficient stimulus to evoke distress calling.[52, 53]

by the mother that include protective behaviors which effectively curtail interaction by others with the infant. Such first-stage behaviors ordinarily would not elicit responses from animals other than the mother, since these might provoke protective aggression by the mother against the intervening animal. Thus, it might be that a different set of signals would be needed, having the added meaning, "Mother is not present," to evoke comfort-producing maternal behavior by other animals. There is some evidence from other species that this might be so; but in this study of pigtails, the depressive reaction in the infants did *not* elicit maternal behaviors by other animals.

It did, however, evoke one striking type of protective intervention that in our laboratory has been most unusual in other contexts: behavior by the adult male. Ordinarily he remains quite aloof from the infants, not getting involved in any social interactions with them and showing only an increased tolerance for their bouncing about in play. Although he will not infrequently intervene to avert or terminate aggressive encounters between adult females, he usually shows no protective behavior toward the infants, which, of course, is generally provided by the mother. However, in this study we noted many protective interventions by the adult male when the motherless, depressed infant became the object of aggression from another animal, adult or infant. These interventions were of two types: (1) the male threatened the aggressive animal; or (2) he placed himself physically between the infant and the aggressive animal. However, the male in no other way provided comfort to the depressed infant.*

*Itani[28] has reported that in the Japanese monkey dominant males show considerable maternal behavior to infants during the breeding season when the mothers are having new babies and ignore their older infants. We have also seen considerable maternal behavior by bonnet males.

Just as the stage-one reaction has time-limited survival value, the depressive reaction, if it continued unabated and indefinitely, would have a host of deleterious effects. This state of minimal action and interaction sharply curtails experience of the outside world and would virtually bring further development to an end. The animal growing physically larger but otherwise not acquiring the personal and social skills of his species would not long survive, or would survive only as an outcast with no possibility of reproductive success.

The consequences of prolongation of this depressive reaction were tragically evident in the babies with the hospitalism syndrome described by Spitz. Not only did further development cease, but there was regression and ultimately no fitness for survival. This illustrates the fact that in the human, evolutionary adaptation has accentuated the role of the mothering figure to an exquisite degree. The human infant cannot survive without a mother. The monkey infant, on the other hand, from an early age, *is* able to survive without mother, and this was illustrated by the third stage of reaction: the recovery from the depression. This, we believe, he was able to do because of his locomotor ability. As Spitz noted, the human infant for a long time has no locomotor ability, and even when he begins to acquire it, he is very much dependent on other figures in the environment to provide the opportunities to use it. The monkey infant, however, after the first few weeks of life is able to move about, and within a relatively short time he has very considerable locomotor ability. This provides the mechanism which enables him on his own to re-engage the environment to find new sources of comfort, with a reasonable likelihood of success not only in surviving but also in resuming development, acquiring new knowledge and new skills, and achieving social growth through interaction with peers and adults.

Neurobiological Considerations

In comparing the reaction in monkeys with that in humans, it is necessary to consider developmental levels. The similarities noted by earlier workers and found also in this study, consisting of a sequence of agitated searching followed in 3 infants by a depressive reaction strikingly similar to the anaclitic depression described by Spitz for young human infants, suggests that in this instance we are dealing with relatively undifferentiated biological response systems which are essentially common to both species. Engel proposed that the central nervous system is organized to mediate two opposite patterns of response to a mounting need. The first, the mobilizing flight-fight reaction, he considers to be the biological anlage of anxiety. The second, the conservation-withdrawal system, which comes into play if and when the first reaction threatens exhaustion, he considers to be the biological anlage of depression-withdrawal. We believe that our data strongly corroborate Engel's theory of the two stages of biological response. Further we believe that contemporary neurophysiological data and theory enable us to understand more fully the neural mechanisms mediating these two organismic responses and their affective components.

Hess[26] had demonstrated the existence in the diencephalon of a trophotrophic, energy-conserving process, working mostly through the parasympathetic nervous system, and an ergotrophic, or mobilizing system, working through the sympathetic division. Still earlier Cannon[8] and Bard[2] had shown the involvement of the diencephalon in emotional expression and feeling. At the same time Papez[39] had indicated that the limbic system was ideally arranged to handle the long-lasting and intense aspects of experience usually associated with emotion. Then, MacLean[36, 37] demonstrated the relationship between the limbic system and the visceral-autonomic systems which have long been implicated in all theories of emotion. Also of importance was the demonstration by Moruzzi and Magoun[38] that the reticular formation mediates a nonspecific activating system intimately related to states of arousal. Now, Riss[46,47] and Pribram[41, 42] in separate proposals offer data and hypotheses which we believe may be applied to clarify the two biological response systems and their associated affects noted by Engel.

Riss, mindful of the explanatory power of Schneirla's[56] theory that approach and withdrawal processes, based on stimulus intensity, underlie behavior, has concluded from his considerations of the evolution of the vertebrate nervous system that there is a comfort-distress dichotomy of whole organism reactions. At the lowest levels of the nervous system, segmental and intersegmental, there are two opposing reflex reactions: moderate stimulation, in general, produces "maintain contact" reflexes and reactions, whereas strong stimulation produces "break contact" reactions. These fundamental dichotomously arranged reactions are each linked at a higher level (up to the primitive forebrain) with the connections for intrinsic responses which are directional, visceral (including endocrinological), vocal, and distinctly expressive. Comfort is associated with "maintain contact" reflexes, and distress with "break contact" reflexes. In lower organisms and in the early developmental stages of higher organisms, these dichotomous reactions are carried out without participation by neothalamus and neocortex, but in the more advanced stages of higher organisms these reactions are considerably modified by the activity of these structures. Riss proposes further that another dichotomous arrangement of the nervous system intersects the first. Facilitation-suppression of the general level of sensorimotor activity is this second dimension. This dimension is mediated via the reticular formation which, through its many connections,

provides an index of average sensorimotor activity. Its rostral projections to the hypothalamus and the paleothalamus (and in turn to the striatum) provide for adjustments in the level of visceral and somatic activity.

The two responses described by Engel are both distress reactions, the first being associated with facilitation and the second with suppression. Facilitated distress is the reaction which tends to overcome or escape from the source of distress, *or* find again the source of comfort. Recovery from the distress reaction in the presence of a distressing environment is thought to be not possible without the intervention of comfort-producing stimulation. Persistence of this high-energy disturbance (facilitated distress) is assumed to lead to changes in the internal milieu (probably chemical) that are essentially similar to the malaise of physical illness, and which induce through the diencephalic level a change in the adjustment from facilitation to suppression of the distress reaction. A collapsed posture, immobility, and a break with the environment ensue. This state of enforced rest in the physically sick organism is the reaction which ordinarily permits subsequent recovery and survival.

To understand better the comfort-distress reactions in the context of the infant-mother relationship, some of Schneirla's formulations[56] may be considered. (In order to focus on the infant's reactions, this discussion will deliberately omit the extremely important aspect of the mother's interest in the infant and the very significant process of reciprocal stimulation.) The neonate is first attracted to the mother through biotaxic effects such as temperature, visual, olfactory, auditory, and gentle contactual stimuli, which produce in the infant a comfort reaction. Presence of such stimuli even in the face of aversive stimuli may maintain the comfort reaction.[49] To begin with (essentially without neothalamus and neocortex), there is not much specificity with respect to the object which provides these stimuli, although the tension relief through feeding and bladder and bowel evacuation begins to organize the percept of the parent. The removal of these biotaxic stimuli usually leads to a distress reaction. In higher organisms, psychotaxic processes arise through reactions to taxic and biotaxic stimuli which become modified through the integration of maturation and experience by virtue of such processes as conditioning, learning, and concept formation (neothalamus and neocortex), introducing plasticity of behavior. Psychosocial types of bond formation then arise, in which the *meanings* rather than the immediate physiological effects of stimuli are functional. In the older infant, then, we may assume that the presence of the "mother" is primarily comfortable and her absence is primarily distressing. In her absence the initial reaction is facilitated distress. When this reaction has lasted long enough to produce changes in the internal milieu, the reaction changes to suppressed distress, or conservation-withdrawal. Just as enforced rest in the physically ill organism admits the possibility of recovery and survival, conservation-withdrawal in the bereft organism allows the recouping of expended internal resources and admits the possibility of recovery and survival through the gradual finding of other sources of comfort. Since the organismic reaction is essentially the same, although the precipitating factors are different, it is easy to understand why Darwin, Spitz, and others commented upon the similarity between depressive illness and physical illness. It also lends strength to Engel's suggestion that grief is a disease.[15]

In these ways the organism deals with a disturbing environment by action on the environment—fight, flight, search, and withdrawal. Sometimes these actions are coordinated in sequence. For example, Chance[10] described a number of acts and

postures in animals which he called "cut-off." They consisted of behaviors which reduce visual stimulation from another animal, especially during agonistic encounters. His interpretation was that the reduced visual stimulation allows the level of arousal to decline, presumably leading to a lowering of fight-or-flight motivation and thus permitting a social relationship to exist. In his view the "cut-off" behaviors buy time for opposing motivation to rise again, and thus provide greater flexibility of behavior in response to stimulation. Similarly, in our behavioral taxonomy for monkeys,[34] we have described a behavior we call "communication avoidance," which also consists primarily of the avoidance of specific visual stimulation.

There is another way that the organism can handle disturbing sensory input to the central nervous system, and that is by way of efferent control of the input itself; this may operate peripherally or centrally. Pribram has developed a theory that the emotional aspects of behavior are based on cerebral efferent control of input. This involves internal processing of input rather than action on the environment, or literally, e-motion. He points out that the word "emotion" comes from the Latin *emovere*, which means to be "out of" or "away from" motion. There are two neurophysiological mechanisms for cerebral efferent control of input: self-inhibition and collateral inhibition of cortical cells. Both self-inhibition and collateral inhibition are ubiquitous, and each process may be either enhanced or inhibited. This requires a fourfold controlling process, and he cites evidence for four mechanisms of cerebrofugal control in which various parts of the neocortex and the primitive forebrain, including the limbic system, and especially the amygdala and the hippocampus, are intimately involved. His theory is based on the processing of incongruity in input, with incongruity being determined by a match between current input and the ongoing neural programs or plans, which are a set of genetic and experiential memory mechanisms that organize the perceptions and behavior of the organism. Input which is sufficiently incongruous can interrupt the ongoing plans; there is a temporary discontinuity, literally e-motion.

Pribram's thesis is that the emotions are those processes which (1) reflect the state of relative organization or disorganization of the ordinarily stable configuration of neural events; and (2) reflect those mechanisms which operate to redress this imbalance, not through action, but by the regulation of input. There is one set of emotions that reflects the concurrent state of order or disruption and a second set which returns the system to control. Regulation is achieved either (1) through the incorporation of input, a "participatory reaction," and the eventual achievement of congruity through alteration in the neural mechanism or model, or (2) through gating-out the incongruous input and preserving old congruities. The first he calls "external control;" it is essentially based on collateral inhibition which reduces redundancy, enhances contrast, and provides opportunity for the registration of the input. Associated affects are satisfying and optimistic—e.g., interest, confidence, hope, and zest. This is the process which allows for progressively more individuated control through the development of repertoires of plans and behaviors. The whole-organism reaction would presumably be comfort. The second mode of regulation he calls "internal control" which is essentially conservative and tends to return the system to the status quo ante. It is based on self-inhibition, which increases redundancy and decreases information density in the cortex. The associated affects, while sometimes gratifying, are essentially pessimistic—e.g., annoyance, apprehension, hopelessness, and depression. The whole-

organism reaction would presumably be distress.

His theory is memory-based, which means that differentiated affects are more and more idiosyncratically and experientially determined. In the young organism, memories are essentially species memories—i.e., primarily genetically programmed plans. He says that for the infant, stability is maintained in large part through the cyclic visceral-autonomic functions of breathing, heart rate, etc., which provide the major sets of inputs to be fashioned into a neuronal model. (An obvious exception would be the environmental processing, largely visual, which takes place during the alert inactive state of the neonate.[64]) Pribram continues, "When this highly redundant pattern is interrupted by an incongruous input, whether a loud noise, pain, or drive stimulus,* then disequilibration takes place. This is accompanied by signs of distress which prompt action by the mother, which returns the infant to its redundant stable base line. Thus reinforcement is initially accomplished by the care-taking object, effecting a return to internal control."

The absence of "mother" (or before that, of her biotaxic stimuli) is presumably an incongruous input (not matching existing plans). The state of disequilibrium is reflected in the concurrent affect of dissatisfaction and the prospective affect of anxiety. Pribram says, "Anxiety occurs when there is ambiguity about subsequent reinforcement. . . . defensive processes then take place in order to achieve internal control." It seems to us, rather, that anxiety is actually a special case of participatory reaction. Ordinarily, participatory reactions are called forth following a match which predicts that through active processing of input a successful outcome (i.e., the achievement of new congruity) is likely. Anxiety may involve an *obligatory* participatory reaction, based not on the likelihood of success via new congruity but on the search for old congruity, in which the environment is scanned for a particular congruous input—e.g., "mother" or her biotaxic stimuli. In our view defensive processes (e.g., repression) might come next, to handle a persistent incongruity and the additional incongruity stemming from the anxiety itself.

Pribram also considers "depression-withdrawal" and says, "Because the efficacy of external control in the infant is so dependent on the action of the care-taking person, the absence of the latter precipitates a predominantly preparatory [defensive] operation. . . . Here the infant seemingly gives up . . . thereby gating out disequilibrating input." We would consider this as the second stage, following anxiety. He considers the differentiated affect, depression, to stem from "the interruption of a shared plan of action, [as after object loss] thereby making the achievement of external control seem unlikely."

In sum, we propose that at an undifferentiated level the removal of the mother's biotaxic stimuli first produces in the infant a facilitated distress reaction, characterized by a high level of motoric, visceral-autonomic, vocal, and expressive activity, which ordinarily functions to regain mother's comfort-producing stimuli, primarily through the communicative value of the reaction, which has evolved through its selective advantage. We presume that it is accompanied by an undifferentiated unpleasant affect of apprehension (the precursor of anxiety) based on uncertain sensory incongruity. If comfort-produc-

*He notes that drive stimuli come from within the organism and cannot be escaped, and that therefore incongruities induced by drive stimuli can only be eliminated by the incorporation of some other input. This is equivalent to saying that distressing inner stimuli may be processed in the presence of the comfort-producing mother. The same would be true also of distressing external stimuli. It should be noted that the mother herself provides a wealthy source of stimuli which call forth participatory reactions.

ing stimuli do not materialize, distress persists but through its catabolic effects suppression occurs. This reaction is characterized by a postural collapse, immobility, withdrawal from the environment, and reduction in visceral-autonomic, vocal, and expressive activity. This too may have communicative value. We presume that it is accompanied by an undifferentiated unpleasant affect of pessimism (the precursor of depression) based on unmanageable sensory incongruity. This conservative reaction preserves the organism's integrity until comfort-producing stimuli may arise. It is suggested that both stages of the distress reaction are mediated by neural mechanisms integrated at midbrain, diencephalic, and paleocortical levels.

With experience, maturation, and learning (and the participation of neothalamus and neocortex) a variety of changes occurs which leads to alterations in the gross response patterns. "Mother" becomes a psychotaxic object. Repertoires of behaviors and plans emerge, providing new coping and controlling mechanisms, including, differentiation of the affects and their utilization as internal communicative signals. In both monkeys and humans, however, we may see in the young infant separated from mother the relatively undifferentiated response patterns, even after the mother has become a psychotaxic object.

Summary and Conclusions

Of possible effects on an infant of separation from mother there seem to be at least seven kinds, although of course they are not mutually exclusive. This classification is artificial to the extent that it excludes from immediate consideration all the other concurrent variables in the situation in an attempt to isolate the infant effects, but thereby it has the virtue of providing for empirical studies of the various effects by manipulation of the other variables. First, there is the effect on the infant at the time of separation. This may be manifested behaviorally, somatically, or both ways (as is true of most of the effects). Second, there is the course of effect if the separation persists. Third, there is the effect on the subsequent development of the infant. Fourth, there is the effect of producing within the infant a disposition to overt disturbance of function in the face of other factors. Fifth, there is the effect on the infant-mother relationship. Sixth, seventh, and eighth, there are the effects that only become manifest upon the mother's return, aside from that on the infant-mother relationship, which may be short-term, have a changing course, or be relatively permanent.

In a study of group-living pigtail monkey infants whose mothers were removed for 4 weeks we found a number of these effects. The infants reacted to separation with agitated searching and calling and a fall in peer-directed behaviors. In 3 of the 4 this was followed in a day by a deep depression similar to the anaclitic depression of human infants described by Spitz. Two of these 3 showed the abnormal behavior of penis sucking. At the end of a week recovery started with a return of interest in and activity involving first, the inanimate environment and then the social environment, especially peers. By the time the mother was returned the infants appeared almost normal in their behavior. The fourth infant, whose mother was the dominant female, showed greater coping ability, succeeded in getting attention from the other adult females, and did not show the marked depression, although she showed many of the same behavioral changes as the others, such as a significant decrease in play and significant increases in self-directed behaviors and inanimate object exploration. Upon reunion there was a dramatic reassertion of the dyadic relationship in all, with the manifestations of increased closeness still evident in the second month follow-

ing the mother's return. As compared to the preseparation period, clinging by the infant, nipple contact, and protective enclosure by the mother all rose, whereas play and inanimate object exploration fell, despite the fact that normally all these behaviors are going in the other direction at these age periods. Also, the infants went shorter distances away from the mother and stayed away for shorter periods. In addition, the mothers showed little or no abdyadic behavior, such as punitive deterrence and nipple withdrawal. In short, the mother-infant relationship became closer as a result of the separation.

There were certain effects that we either did not see or did not study. We saw no effects following reunion other than that on the dyadic relationship. There was no obvious interference with subsequent development, although this requires further study.

We have not yet studied the possibility that separation creates a disposition to later disturbance. We have seen 1 infant who had had a depression upon separation from his mother, who showed a milder depression and the abnormal behavior of penis sucking upon the birth of a sibling. This problem needs further study, especially with respect to the development of somatic disturbance, since we know from humans that object loss is so importantly involved in psychosomatic illness. In this regard we saw no obvious somatic illness, physical health having been generally sustained.

The various infant responses and their communicative aspects were related to the hierarchical regulation of monkey behavior. When the phases of reaction were considered in terms of survival value and compared with those of humans, with emphasis on developmental levels and achievements, it was concluded that monkey infants have a greater chance of survival without a mother figure because of their greater locomotor ability.

In line with the thesis that much of behavior is organized around a comfort-distress dimension, and that the mother's presence is primarily comforting while her absence is distressing, the reaction to separation is seen as a distress reaction. Its stages appear to represent successive efforts at adaptation based on available response systems, evolved for their selective advantage, or developed ontogenetically, especially through dominance-hierarchical regulatory influences; at an undifferentiated level these response systems seem common to man and monkey. This is in line with an earlier theory proposed by Engel. The first stage of agitated distress is thought to be an ergotrophic response calculated (by evolutionary adaptation) to regain comfort or remove distress. The subsequent stage of persistent distress, withdrawal, is considered a trophotrophic response calculated to conserve energy and avoid injury until comfort may be regained. The emotional aspects of these responses, both phenomenologically and in their communicated effect upon human observers, appear to be related to anxiety and depression.

The neural mechanisms underlying these responses and their emotional aspects are discussed in the light of recent proposals by Riss on the organization of the nervous system to mediate comfort and distress reactions, and by Pribram on the neurophysiological processes associated with emotion.

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